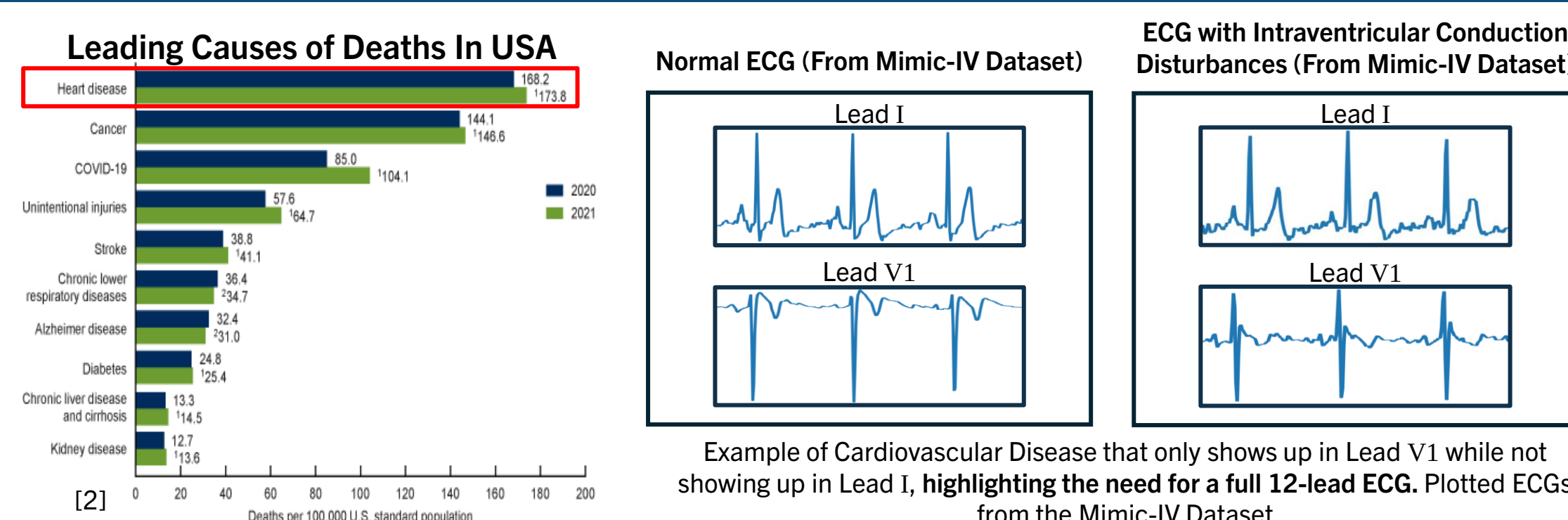


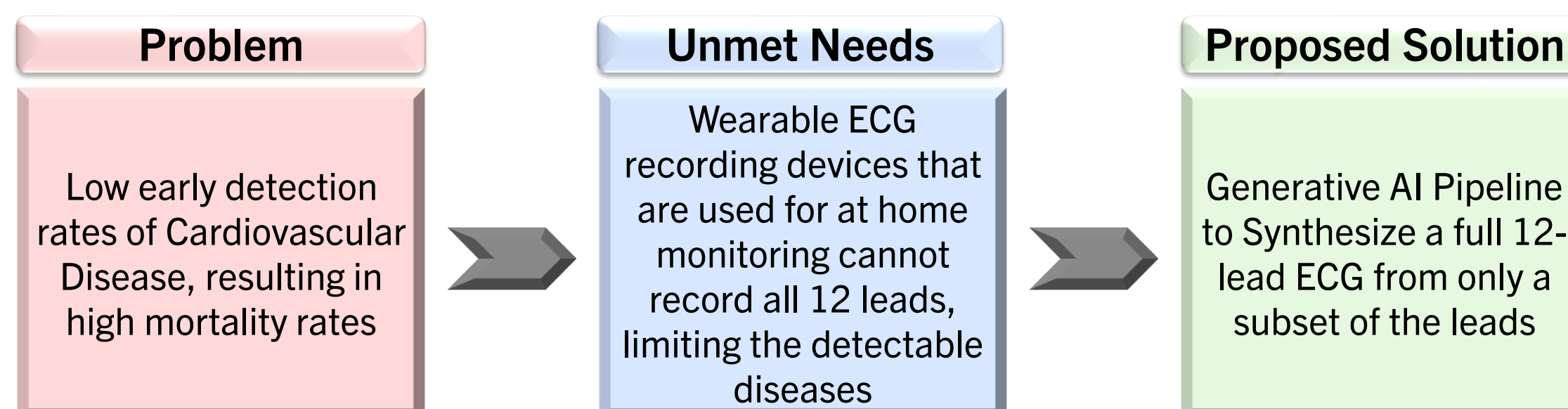
Background



Cardiovascular Disease (CVD) is the leading cause of death in the USA

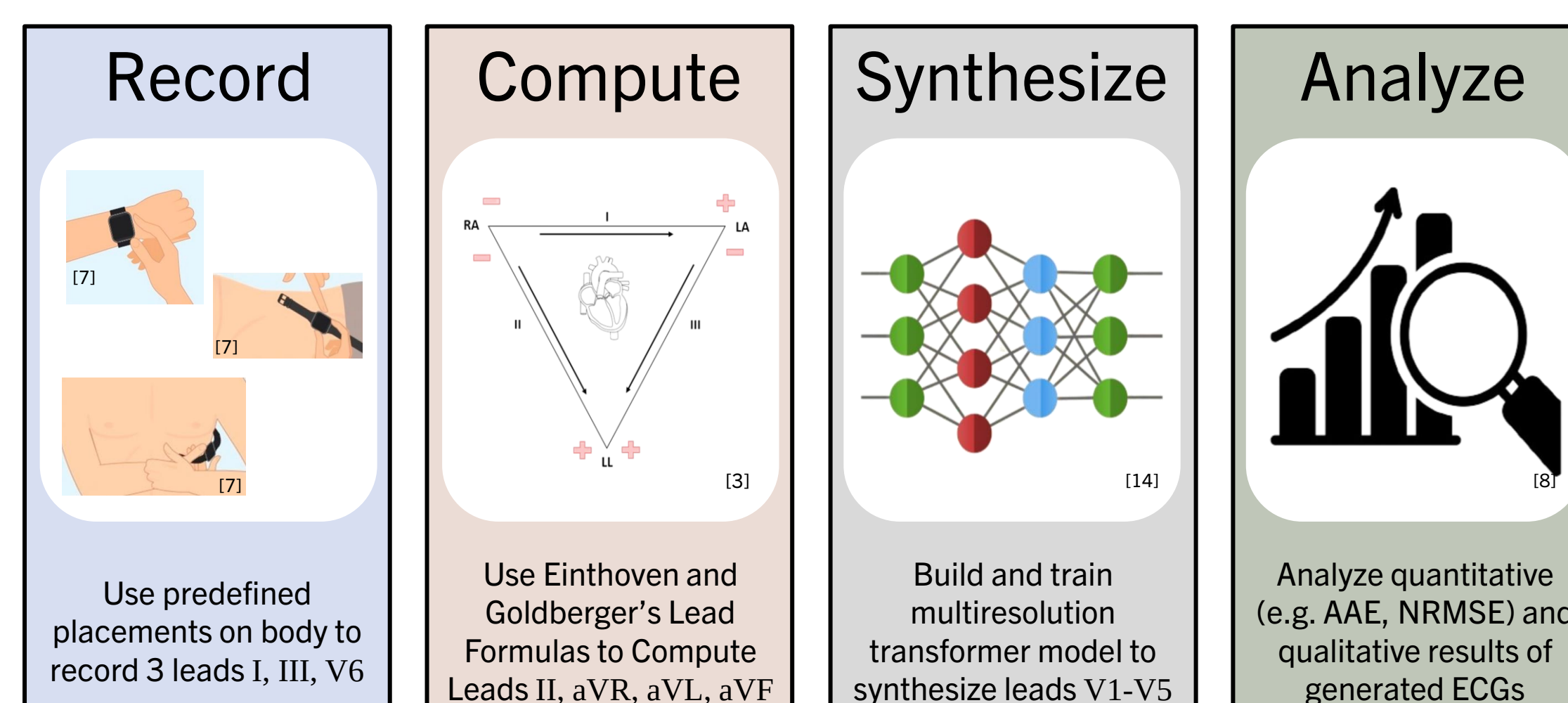
- At home monitoring is crucial for detecting life threatening diseases like Sudden Cardiac Arrest
- Cardiovascular Disease (CVD) can show up only weeks in advance
- Currently ONLY Atrial Fibrillation and Atrial Flutter can be detected from a standard ECG recorded from a wearable device
 - Ventricular diseases, pericarditis and half of all types of Myocardial Infarction CANNOT be detected
- Asymptomatic individuals with CVD rarely get a full 12-lead ECG taken, even at annual check-ups
- Access to 12-Lead ECGs in low and middle-income countries could be as low as 10%

Problem Statement



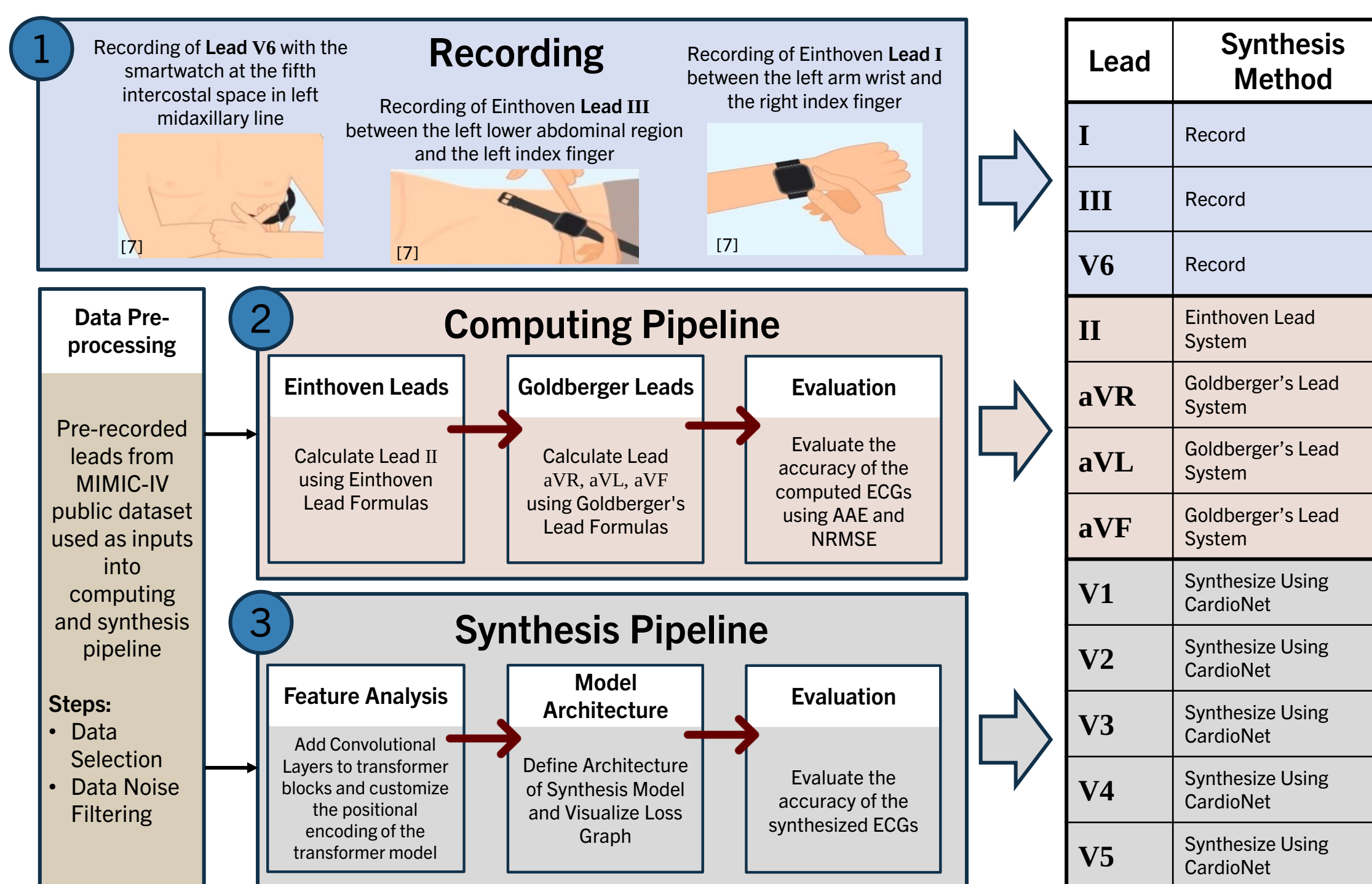
Claim	Impact	Goal
- CardioNet will be able to synthesize a 12-Lead ECG using information from only 3 leads that can be recorded from a smart device - Generated 12-lead ECG is accurate for both normal and abnormal ECGs	- At home monitoring of heart health with only wearable device and CardioNet - Early detection via at home monitoring of CVD can mitigate up to 80% of fatalities	- Create a machine learning pipeline to synthesize a 12-lead ECG from 3 recorded leads - Synthesized ECGs capture both local and global dependencies of ECGs

4-Step Workflow

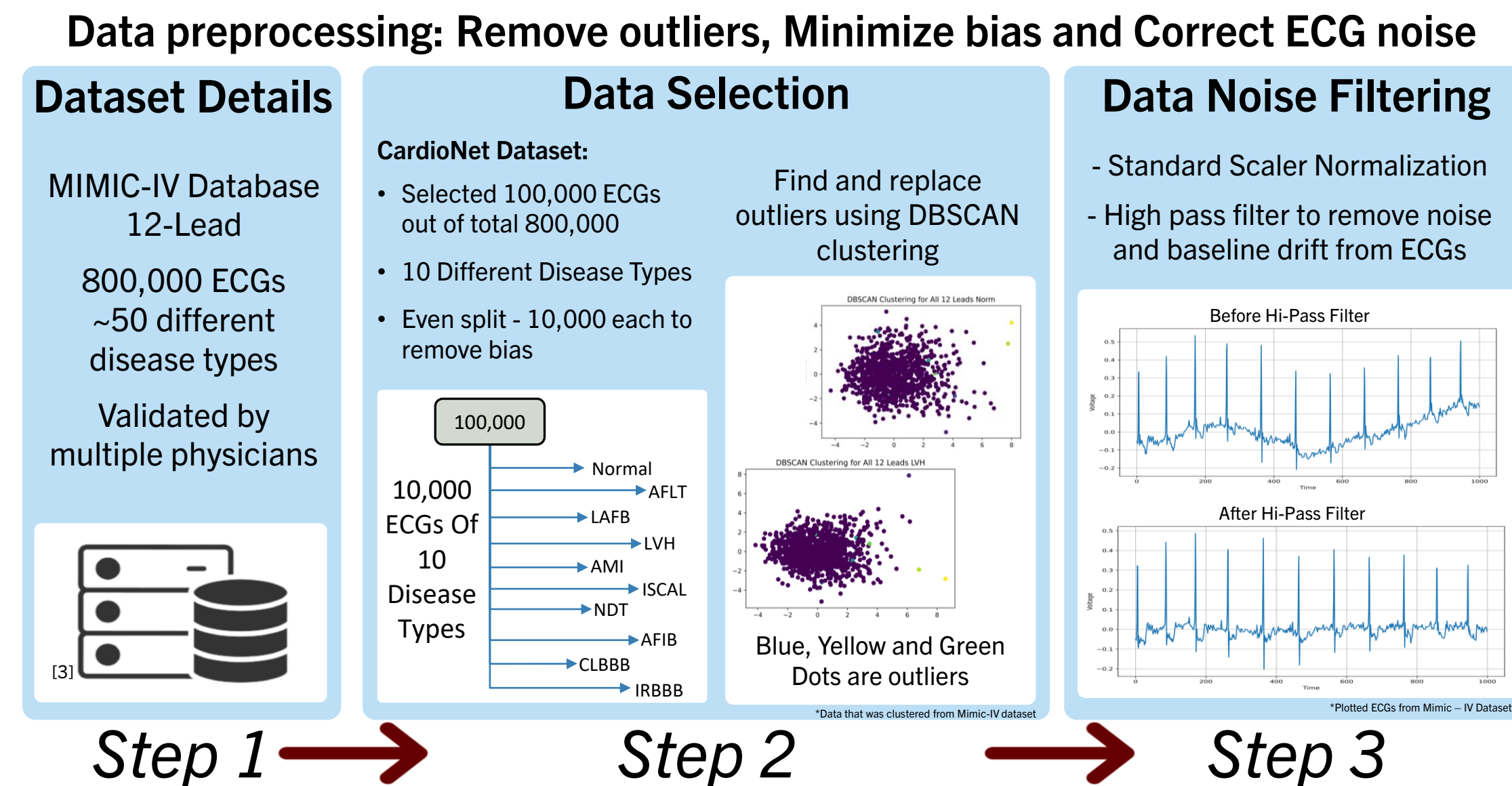


Step 1 → Step 2 → Step 3 → Step 4

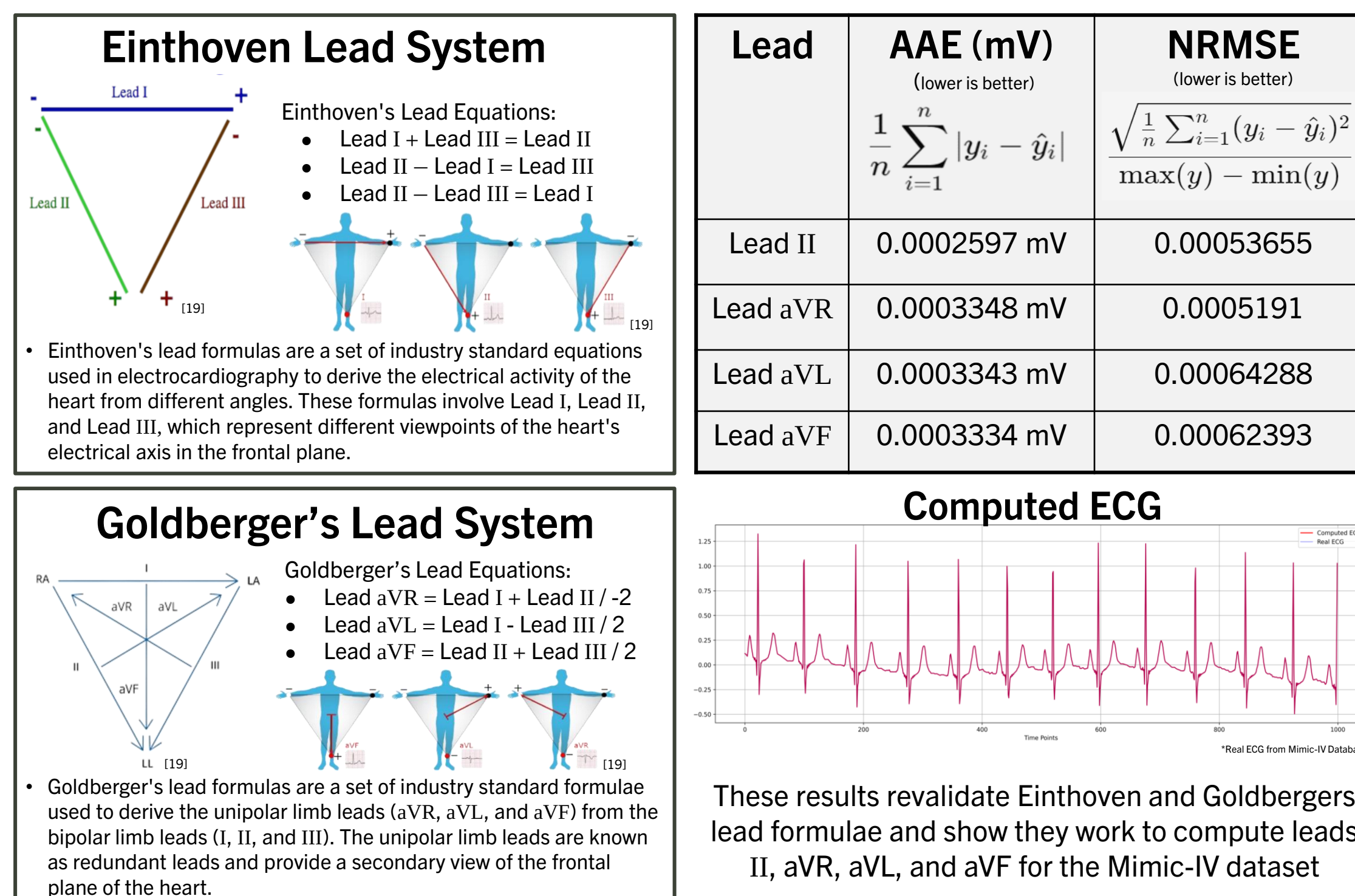
12-Lead ECG Generation Strategy



Data Preprocessing



Computation of Leads (II, aVR, aVL, aVF)

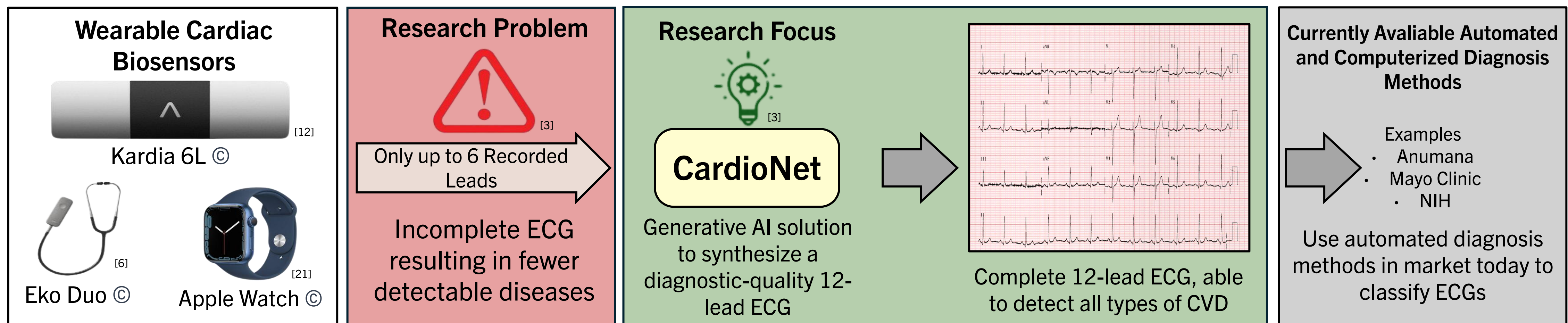


CardioNet: Synthesis of a Diagnostic-Caliber 12-Lead Electrocardiogram Using Generative Neural Networks and Novel Feature Extraction Techniques

Tanay Chitlur – Westview High School Portland OR, USA

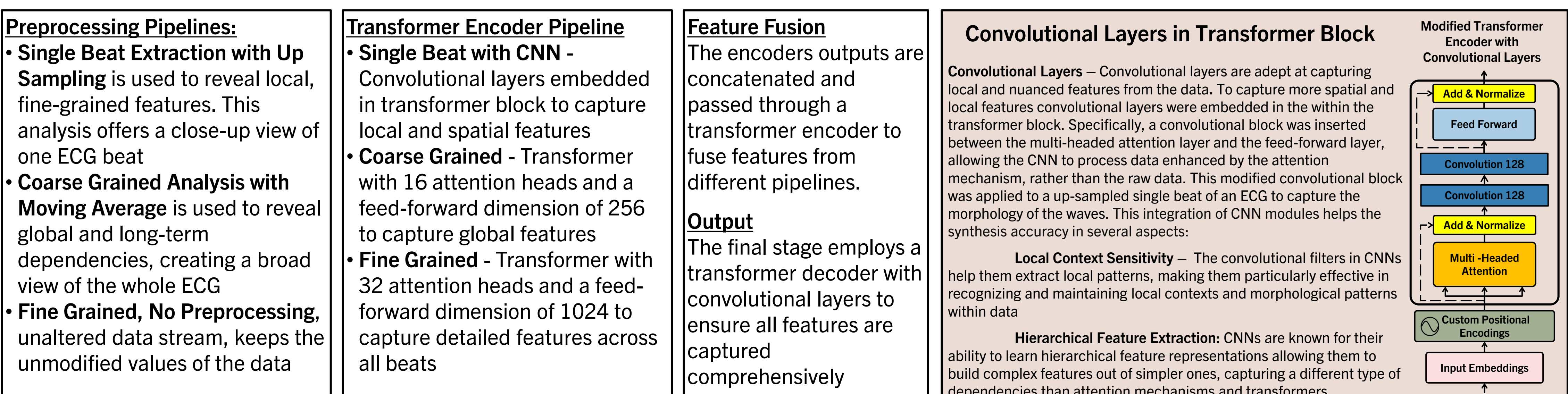
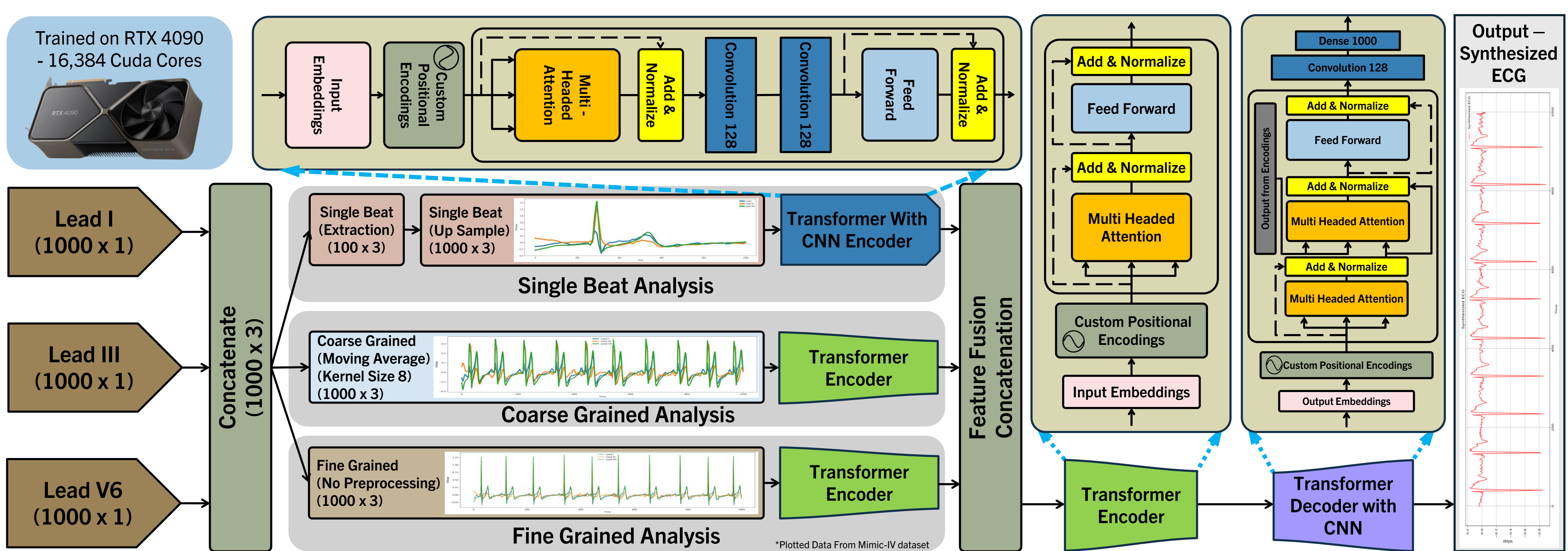
Project ID: ROBO023

Research Focus



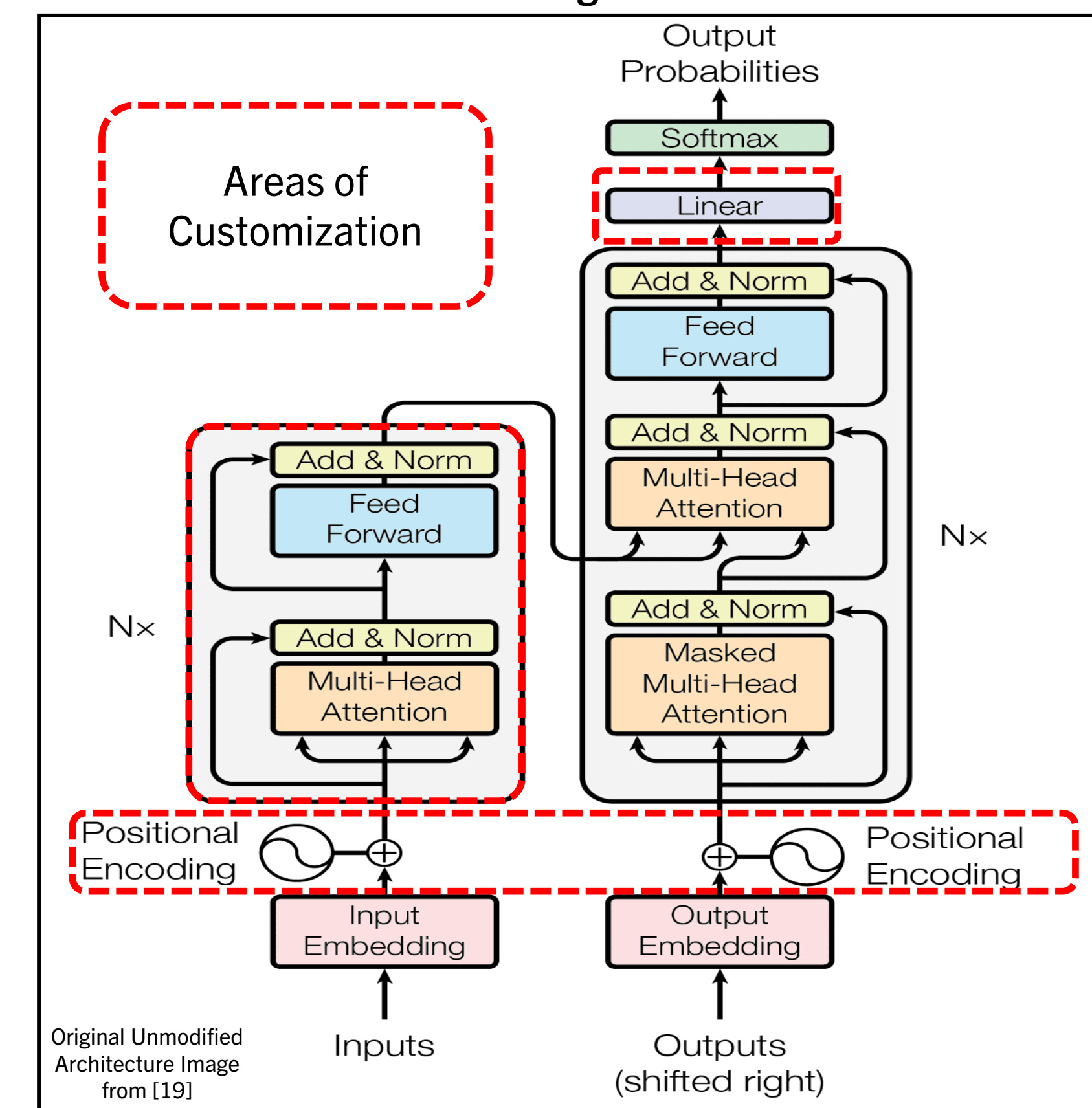
Cardiovascular disease (CVD) is the leading cause of death in the United States. Early detection and at home monitoring can prevent up to 80% of associated fatalities. Smart wearable devices are ideal for at-home monitoring but only record three leads, limiting detectable diseases. This project aims to create a generative AI solution to synthesize a diagnostic-quality 12-lead ECG, necessary for comprehensive CVD detection.

CardioNet: Multiresolution Transformer Network For ECG Synthesis (V1-V5)



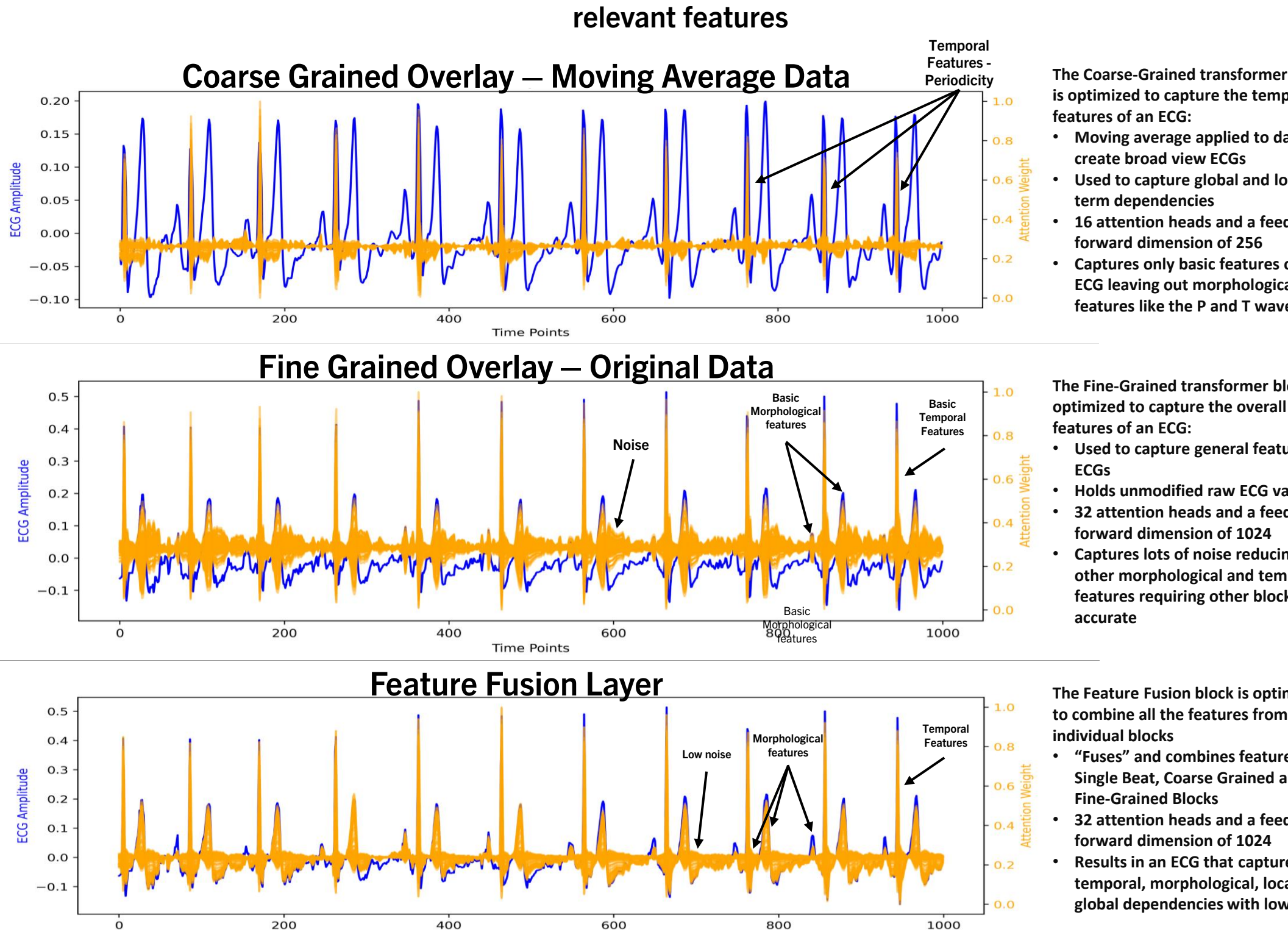
Transformer Architecture Customization

Areas of customization in the general transformer network



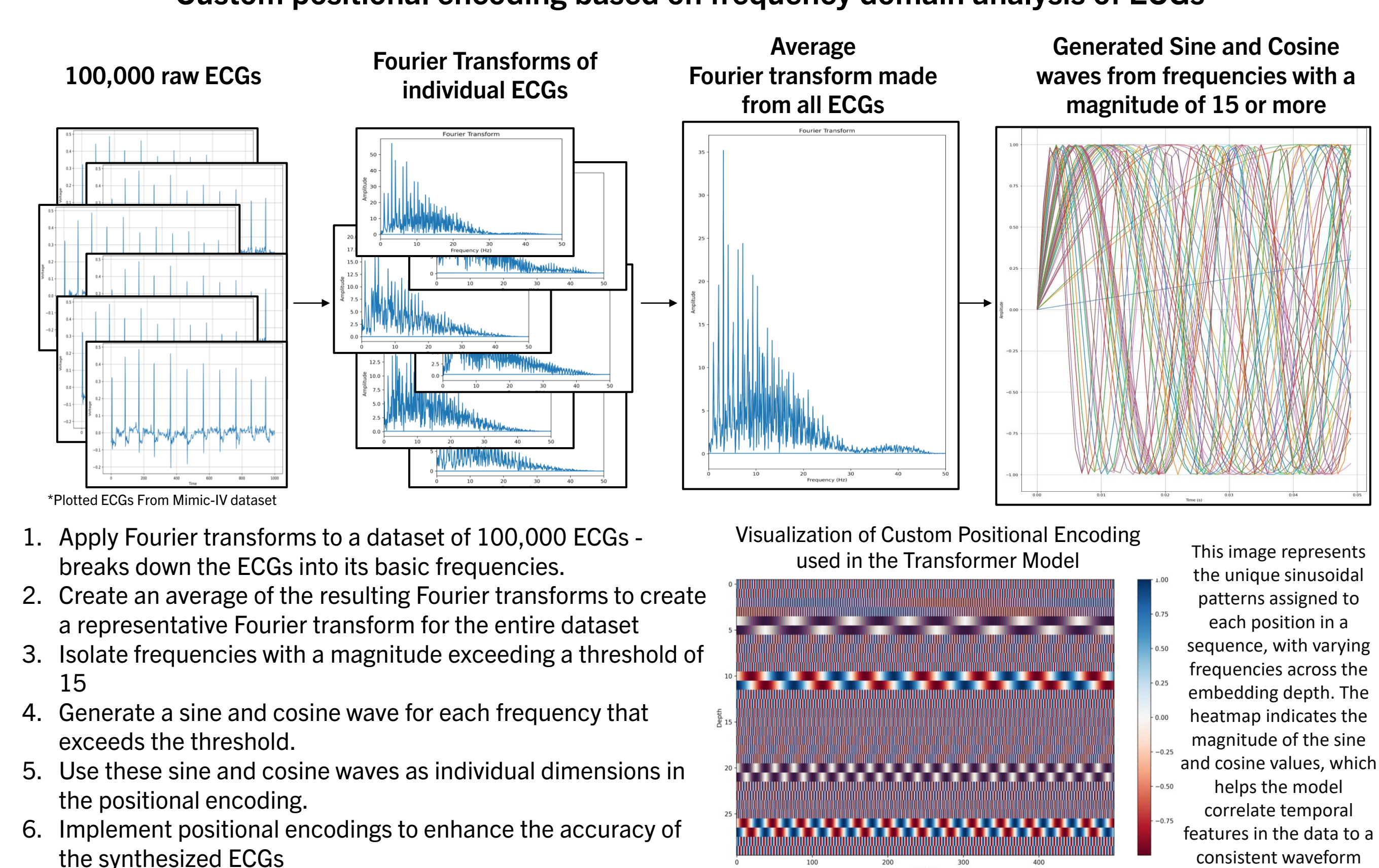
Part 2 – Transformer Attention Overlays

Attention overlay graphs demonstrating how specialized transformer blocks selectively focus on relevant features



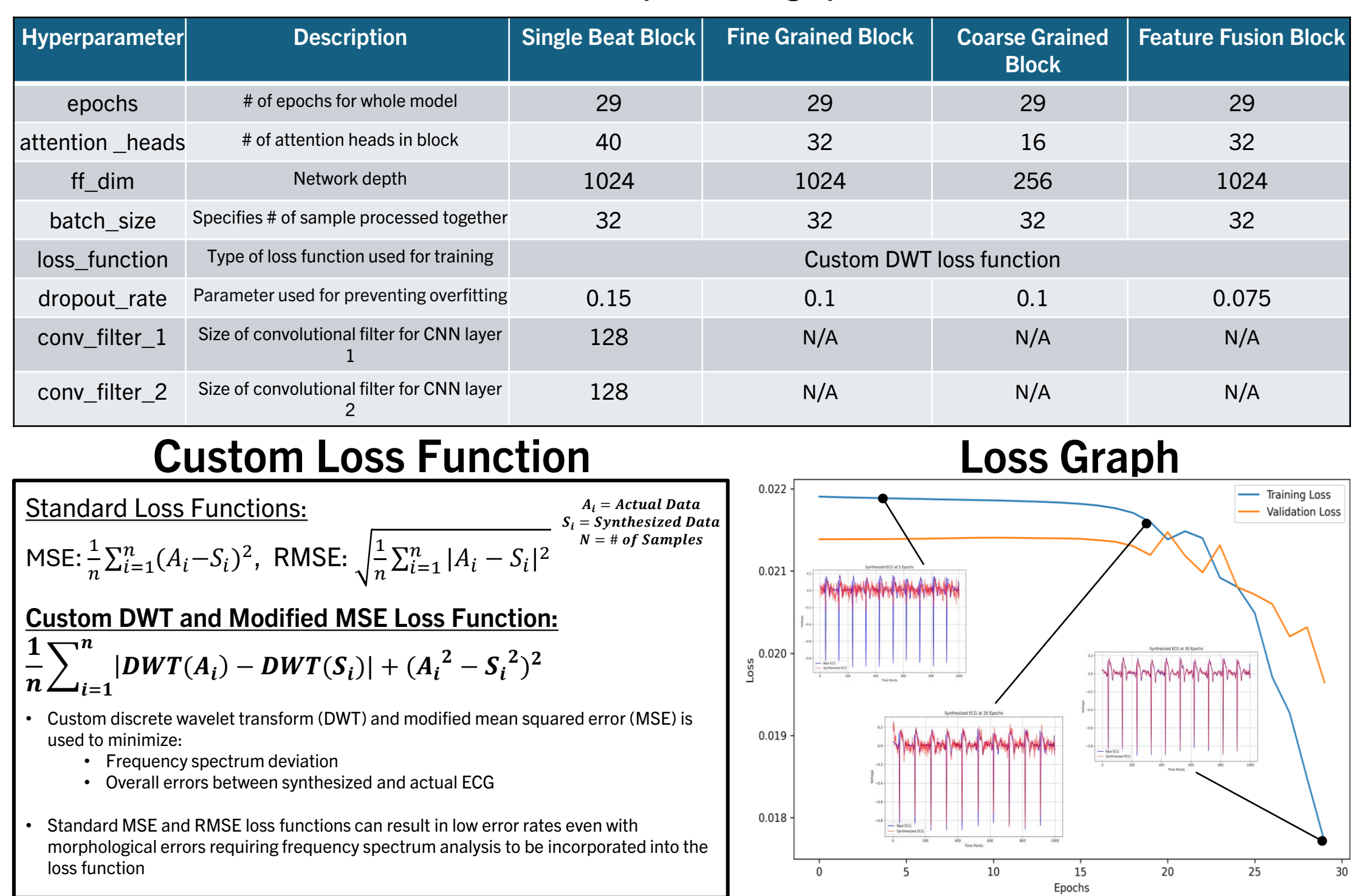
Part 1 – Positional Encoding

Custom positional encoding based on frequency domain analysis of ECGs



Part 3 - Model Training

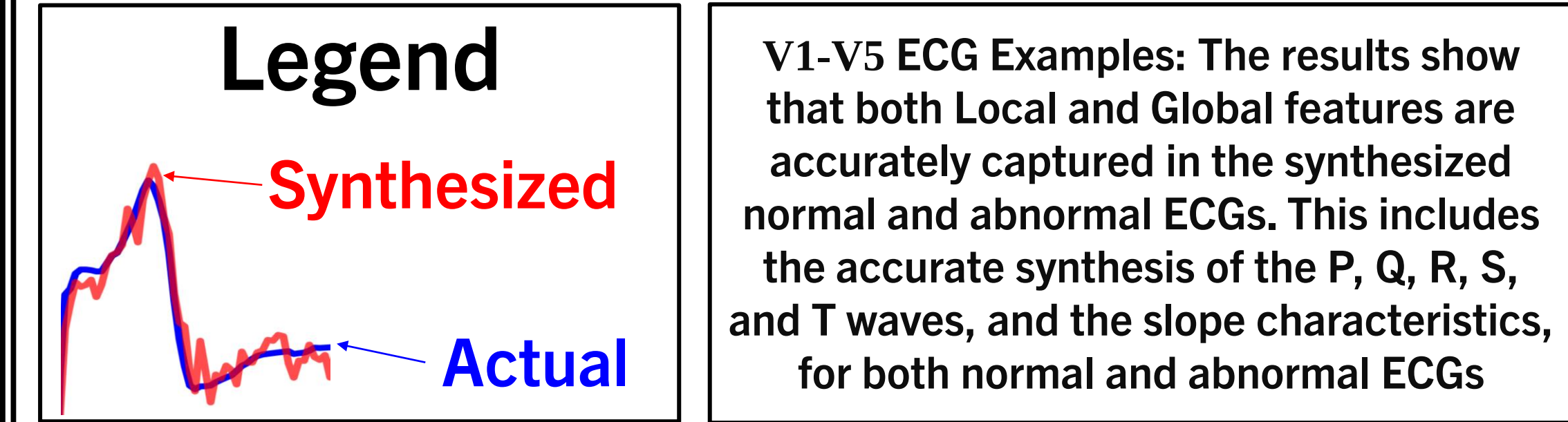
Optimized model hyperparameters and a customized loss function result in an accurate synthesis model with optimal loss graph



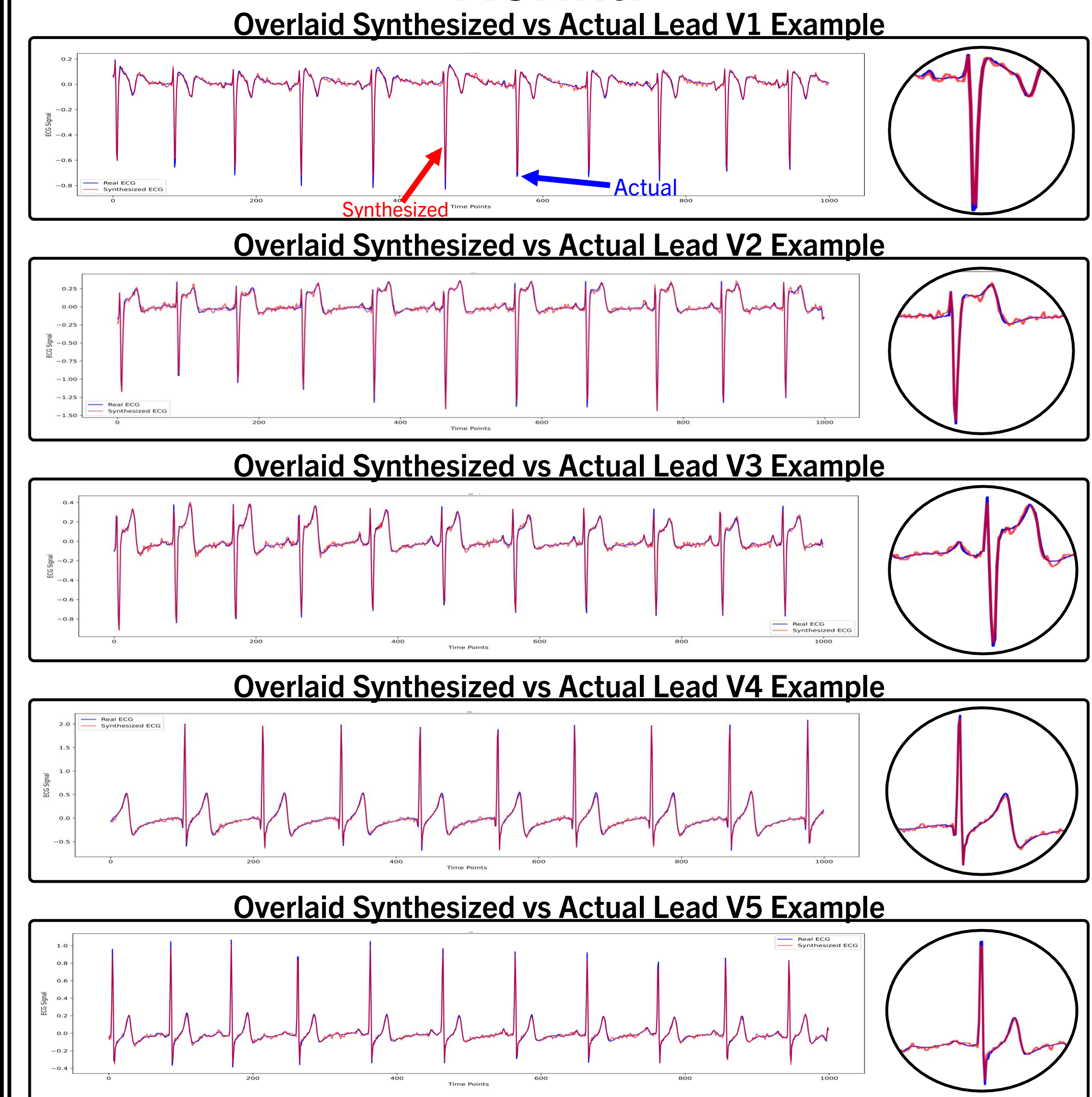
Results

Quality of Results When Tested With 30,000 ECGs: Synthesized Lead vs Actual for Normal and Abnormal ECGs

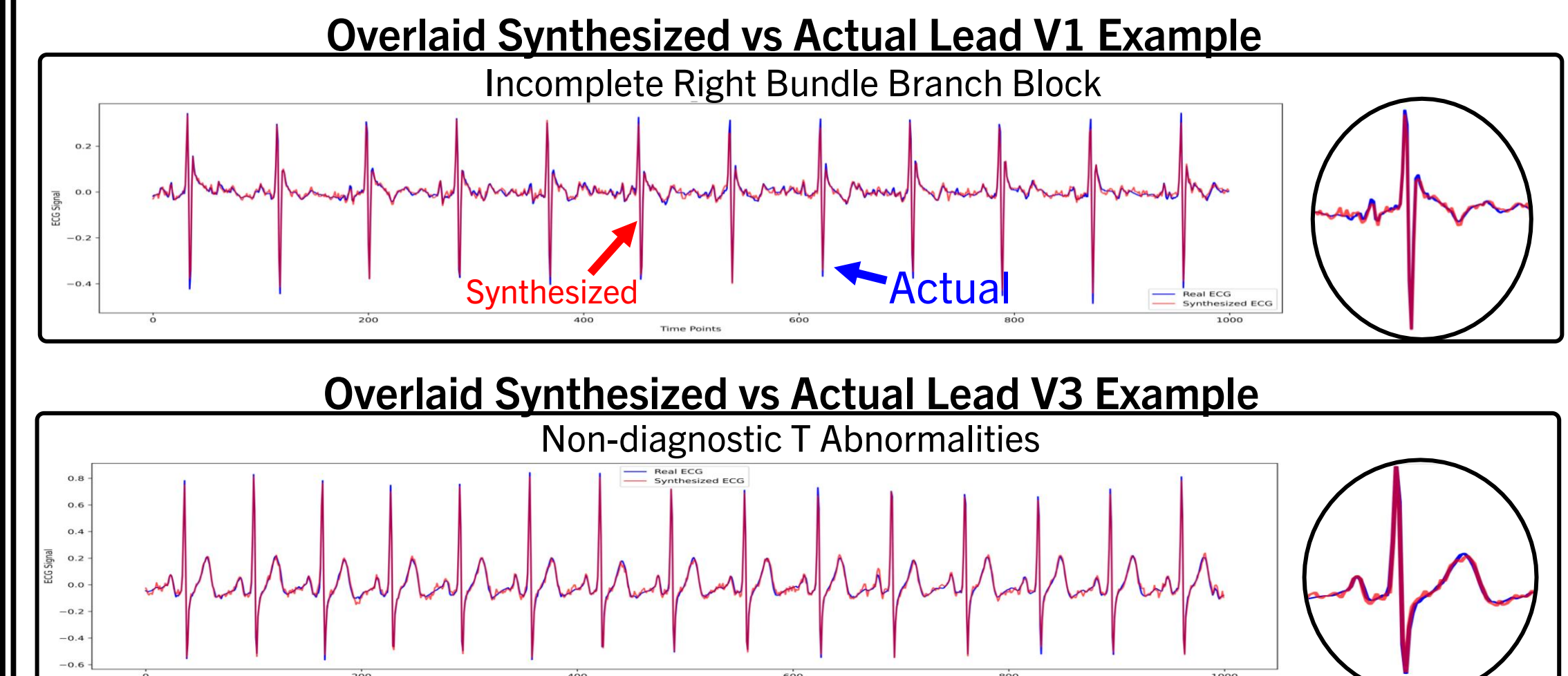
Lead	Average Absolute Error (AAE)	Normalized Root Mean Squared Error (NRMSE)
Lead V1	0.01090 mV	0.0001353
Lead V2	0.01905 mV	0.0001486
Lead V3	0.01849 mV	0.0001338
Lead V4	0.01804 mV	0.0001158
Lead V5	0.01562 mV	0.0001184



Normal



Abnormal



Conclusions

Highlights

- CardioNet is a comprehensive generative pipeline that uses machine learning to create a diagnostic quality 12-lead ECG from 3 recorded leads
- Novel use of multiresolution (single-beat, fine-grained and coarse-grained) transformer to comprehensively capture both global and local dependencies in an ECG
- Use of convolutional neural network layers embedded in single beat transformer block to capture detailed morphological features of an ECG
- Synthesized output ECGs were extensively evaluated to be diagnostic caliber based on quantitative results (low AAE & NRMSE) and qualitative results (morphology analysis)

Future Work

- Use past clinically recorded ECGs along with transfer learning (i.e., targeted retraining of base model) for personalization
 - Incorporate individual metadata (e.g., age, gender, weight) to personalize the model
 - Apply model to all types of cardiovascular disease
- Impact**
- Allows advanced CVD detection using wearable devices - wearable devices can be used to generate complete 12-lead diagnostic quality ECGs
 - Increases early detection rates of cardiovascular disease - enables at home monitoring of CVD
 - Demonstrates effectiveness of generative AI healthcare - novel multiresolution transformer can be applied for the generation of other electrophysiological signals (e.g. EEGs)

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